

YE ZHU (SHE/HER)

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RESEARCH AREAS

Main Research: Generative Models, Computer Vision, Multimodal Learning

Interdisciplinary: Machine Learning for Astrophysics

PROFESSIONAL APPOINTMENT

École Polytechnique, Institut Polytechnique de Paris (IPP), France September 2025 - Now
Monge Tenure-Track Assistant Professor in Computer Science (Laboratoire d'Informatique, LIX).

Princeton University, USA September 2023 - August 2025
Postdoctoral Researcher in Computer Science.
Advisor: Prof. Olga Russakovsky.

EDUCATION

Illinois Institute of Technology, USA August 2023
Ph.D. in Computer Science.
Thesis: Multimodal Learning and Generation - Toward a Multisensory and Creative AI System. (Award for Excellence in Dissertation)
Advisor: Prof. Yan Yan.

Princeton University, USA September 2022 - June 2023
Visiting Ph.D. in Computer Science.
Advisor: Prof. Olga Russakovsky.

Shanghai Jiao Tong University (SJTU), China March 2019
M.S. in Mechanical Engineering.
French Engineering Diploma.

École Polytechnique, France March 2017
French Engineering-track Student.

Shanghai Jiao Tong University (SJTU), China August 2016
B.S. in Mechanical and Automation.
Pre-enrollment before Chinese College Entrance Examination.

RESEARCH PUBLICATIONS

Peer-Reviewed Conference and Journal Publications (* for equal contributions)

[1] **Ye Zhu**, Duo Xu, Zhiwei Deng, Jonathan Tan, Olga Russakovsky. Dynamic Diffusion Schrödinger Bridge in Astrophysical Observational Inversions, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2025. [[Paper](#)] [[Code](#)]

[2] Chi Zuo, Martin B. Møller, Pablo Martínez-Nuevo, Huayang Huang, Yu Wu, **Ye Zhu**. BNMUSIC: Blending Environmental Noises into Personalized Music, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2025. [[Paper](#)] [[Code](#)]

- [3] Ruoyu Wang, Huayang Huang, **Ye Zhu**, Olga Russakovsky, Yu Wu. The Silent Assistant: NoiseQuery as Implicit Guidance for Goal-Driven Image Generation, in *International Conference on Computer Vision (ICCV Highlight)*, 2025. [[Paper](#)] [[Code](#)]
- [4] Yongqi Yang*, Zhihao Qian*, **Ye Zhu**, Olga Russakovsky, and Yu Wu. D³: Scaling Up Deepfake Detection by Learning from Discrepancy, in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. [[Paper](#)] [[Code](#)]
- [5] Duo Xu, Jenna Karcheski, Chi-Yan Law, **Ye Zhu**, Chia-Jung Hsu, and Jonathan Tan. Exploring Magnetic Fields in Molecular Clouds through Denoising Diffusion Probabilistic Models, in *The Astrophysics Journal (APJ)*, 2025. [[Paper](#)] [[Code](#)]
- [6] **Ye Zhu**, Yu Wu, Nicu Sebe, and Yan Yan. Vision + X: A Survey on Multimodal Learning in the Light of Data. in *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2024. [[Paper](#)]
- [7] William Yang, **Ye Zhu**, Zhiwei Deng, Olga Russakovsky. What is Dataset Distillation Learning?, in *International Conference on Machine Learning (ICML)*, 2024. [[Paper](#)] [[Code](#)]
- [8] Ruoyu Wang*, Yongqi Yang*, Zhihao Qian, **Ye Zhu**, and Yu Wu. Diffusion in Diffusion: Cyclic One-Way Diffusion for Text-Vision-Conditioned Generation, in *International Conference on Learning Representations (ICLR)*, 2024. [[Paper](#)] [[Code](#)] [[Project](#)]
- [9] Duo Xu, **Ye Zhu**. Surveying Image Segmentation Approaches in Astronomy. in *Astronomy and Computing*, 2024. [[Paper](#)] (Invited Paper)
- [10] **Ye Zhu***, Zhenhao Zhao*, Xiaoguang Zhu, Yuzhang Shang, and Yan Yan. Supplementing Missing Visions via Dialog for Scene Graph Generations, in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024. [[Paper](#)] [[Code](#)]
- [11] Bin Duan, Hao Tang, Changchang Sun, **Ye Zhu**, and Yan Yan. Mining and Unifying Heterogeneous Contrastive Relations for Weakly-Supervised Actor-Action Segmentation, in *Winter Conference on Applications of Computer Vision (WACV)*, 2024. [[Paper](#)]
- [12] **Ye Zhu**, Yu Wu, Zhiwei Deng, Olga Russakovsky, and Yan Yan. Boundary Guided Learning-Free Semantic Control with Diffusion Models, in *Conference on Neural Information Processing Systems (NeurIPS)*, 2023. [[Paper](#)] [[Code](#)] [[Project](#)] [[Hugging face](#)]
- [13] Duo Xu, Jonathan Tan, Chia-Jung Hsu, and **Ye Zhu**. Denoising Diffusion Probabilistic Models to Predict the Density of Molecular Clouds, in *The Astrophysics Journal (APJ)*, 2023. [[Paper](#)]
- [14] **Ye Zhu**, Yu Wu, Kyle Olszewski, Jian Ren, Sergey Tulyakov, and Yan Yan. Discrete Contrastive Diffusion for Cross-Modal Music and Image Generation, in *International Conference on Learning Representations (ICLR)*, 2023. [[Paper](#)] [[Code](#)] [[Project](#)]
- [15] **Ye Zhu**, Kyle Olszewski, Yu Wu, Panos Achlioptas, Menglei Chai, Yan Yan, and Sergey Tulyakov. Quantized GAN for Complex Music Generation from Dance Videos, in *European Conference on Computer Vision (ECCV)*, 2022. [[Paper](#)] [[Code](#)] [[Project](#)]
- [16] **Ye Zhu**, Yu Wu, Yi Yang, and Yan Yan. Saying the Unseen: Video Descriptions via Dialog Agents, in *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2022. [[Paper](#)] [[Code](#)]
- [17] **Ye Zhu**, Yu Wu, Hugo Latapie, Yi Yang, and Yan Yan. Learning Audio-Visual Correlations From Variational Cross-Modal Generations, in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021. [[Paper](#)] [[Code](#)]
- [18] Xiaoguang Zhu, **Ye Zhu**, Haoyu Wang, Honglin Wen, Yan Yan, and Peilin Liu. Skeleton Sequence and RGB Frame Based Multi-Modality Feature Fusion Network for Action Recognition, in *ACM Transactions on Multimedia Computing Communications and Applications (TOMM)*, 2021. [[Paper](#)]
- [19] **Ye Zhu**, Yu Wu, Yi Yang, and Yan Yan. Describing Unseen Videos via Multi-Modal Cooperative Dialog Agents, in *European Conference on Computer Vision (ECCV)*, 2020. [[Paper](#)] [[Code](#)]

Peer-Reviewed Workshop Publications

- [1] **Ye Zhu**, Yu Wu, Duo Xu, Zhiwei Deng, Yan Yan, and Olga Russakovsky. Discovery and Expansion of New Domains within Diffusion Models, in *Conference on Neural Information Processing Systems, Frontiers in Probabilistic Inference Workshop (NeurIPS Workshop)*, 2025. [\[Paper\]](#) [\[Code\]](#)
- [2] Sai Wang*, **Ye Zhu***, Ruoyu Wang, Amaya Dharmasiri, Olga Russakovsky, and Yu Wu. DETER: Detecting Edited Regions for Detering Generative Manipulations, in *CVPR Responsible Data Workshop (CVPR Workshop Spotlight)*, 2024. [\[Paper\]](#) [\[Project\]](#)
- [3] Matthew Coleman, Olga Russakovsky, Christine Allen-Blanchette, and **Ye Zhu**. Discrete Diffusion Reward Guidance Methods for Offline Reinforcement Learning, in *International Conference on Machine Learning, Sampling and Optimization in Discrete Space (SODS) Workshop (ICML Workshop)*, 2023. [\[Paper\]](#)
- [4] Duo Xu, Jonathan Tan, Chia-Jung Hsu, and **Ye Zhu**. Denoising Diffusion Probabilistic Models to Predict the Number Density of Molecular Clouds in Astronomy, in *International Conference on Learning Representations Physics4ML Workshop (ICLR Workshop)*, 2023. [\[Paper\]](#)
- [5] **Ye Zhu**, Yan Yan, and Oleg Komogortsev. Hierarchical HMM for Eye Movement Classification, in *European Conference on Computer Vision Workshop (ECCV Workshop)*, 2020. [\[Paper\]](#)

Other Preprints

- [1] Yuhan Pei*, Ruoyu Wang*, Yongqi Yang, **Ye Zhu**, Olga Russakovsky, Yu Wu. SOWing Information: Cultivating Contextual Coherence with MLLMs in Image Generation. (arXiv preprint, arXiv:2411.19182), 2024. [\[Paper\]](#) [\[Project\]](#)

TEACHING

AI4ALL Program, Princeton University, USA - Instructor for general lectures and the computer vision project.	July 2024
COS429 Computer Vision, Princeton University, USA - Guest Lecturer on the Generative Models.	Fall 2023
AI4ALL Program, Princeton University, USA - Instructor for general lectures and the NLP project. - Media Coverage on AI4ALL Princeton	July 2023

SELECTED HONORS AND AWARDS

Top Reviewer Award for NeurIPS	2024
EECS Rising Stars , MIT, Cambridge, USA	2024
NeurIPS 2023 Scholar Award, New Orleans, USA	2023
ICCV 2023 DEI Grant, Paris, France	2023
ICLR 2023 Financial Assistance Award, Kigali, Rwanda	2023
ACM-Women Scholarship [Coverage]	2023
Award for Excellence in Dissertation Research for the College of Computing, Illinois Tech, USA	2022

SELECTED PUBLIC TALKS

Columbia University, New York City, USA - Talk on <i>Controllability of Dynamic Generative Models with Theoretical Groundings</i>	July, 2025
DAML Seminar, École Polytechnique, Paris, France (Remote) - Talk on <i>A Sustainable Vision for GenAI through the Lens of Dynamic Generative Models</i>	February, 2025
AI Lab, Princeton, USA - Spotlight talk on <i>Generative AI beyond Scaling</i>	February, 2025
NYC Computer Vision Day, New York City, USA - Lightning talk on <i>Generative Dynamics for Image Controlling and Astrophysical Modeling</i>	February, 2025

Vision Lab, Yale University, New Haven, USA

November, 2024

- Invited talk on *A Sustainable Vision for Generative AI*

TTIC Summer Workshop on Multimodal Artificial Intelligence, Chicago, USA

August, 2024

- Invited talk on *Taming Multimodal Generations via Fundamental Inspirations from Mathematics and Physics*

CVPR Responsible Data Workshop, Seattle, USA

June, 2024

- *DETER: Detecting Edited Regions for Deterring Generative Manipulations*

Talking to Machines Workshop, Riga, Latvia (Remote)

May, 2024

- *GenAI as Content Creators and Beyond*

Immersive Computing Lab, New York University, New York, USA

April, 2024

- *Mining the Latent: A Tuning-Free Paradigm for Versatile Applications with Diffusion Models*

MaVi Group, University of Bristol, UK (Remote)

February, 2024

- *Mining the Latent: A Tuning-Free Paradigm for Versatile Applications with Diffusion Models*

PIXL talk, Princeton University, USA

April 2023

- Topic on diffusion generative models, ML4Astrophysics.

PROFESSIONAL SERVICE

Workshop Organizers

CVPR 2025 **CV4Science workshop** (Workshops on Computer Vision for Science),

CVPR 2024-2025 **ReGenAI Workshop** (Workshops on **R**esponsible **G**enerative **A**I)

Conference Reviewers

Machine Learning and Statistics: NeurIPS 2023-2025, ICLR 2024-2026, ICML 2023-2025, AISTATS 2025-2026, AAAI 2023-2024

Computer Vision and Graphics: CVPR 2022-2025, ECCV 2022-2024, ICCV 2023-2025, WACV 2023-2024, ACM-MM 2021-2022, SIGGRAPH 2024

Signal Processing: ICASSP 2022

Journal Reviewers

Transactions on Machine Learning Research (TMLR), IEEE Transactions on Image Processing (TIP),

IEEE Transactions on Multimedia (TMM), Computer Vision and Image Understanding (CUIV)

EARLY INDUSTRIAL EXPERIENCE

Snap Inc., CA, USA

May 2021 - August 2021

Research intern in Computer Vision, advised by Dr. Kyle Olszewski.

Bang & Olufsen, Struer, Denmark

July 2018 - December 2018

Research intern in Computer Vision, advised by Dr. Sven Ewan Shepstone and Dr. Pablo Martinez-Nuevo.

LINGUISTIC SKILLS AND OTHERS

Chinese (Native), **French** (Professional, DALF & TCF C1 Diploma), **English** (Professional)